CSE 427- Machine Learning

Explainable Medical Cost Prediction System

Using Machine Learning and SHapley Additive exPlanations (SHAP)

Topic Name:

Explainable Medical Cost Prediction System Using Machine Learning and SHapley Additive exPlanations (SHAP)

Meet the Members:

1. Injam Ul Haque
2. Zafra Musarrat Islam
3. Karib Fiyaz

Institution:

BRAC University





Problem Statement

- Medical cost prediction is crucial for healthcare planning
- Traditional models fail to capture complex relationships
- Machine Learning models improve accuracy but:
  - Lack transparency (Black-box problem)

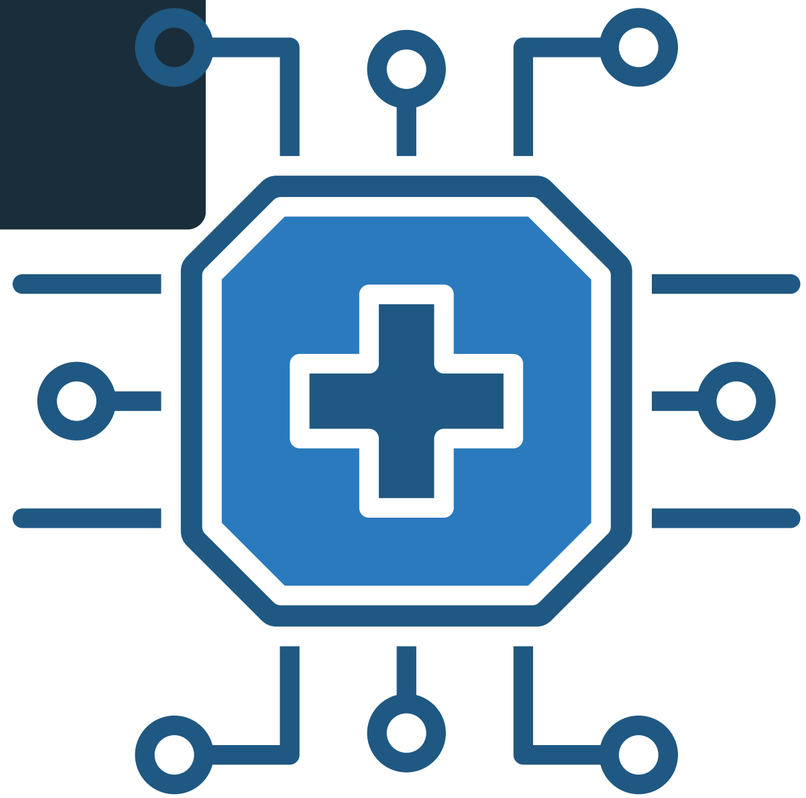
Solution:

Need a system that is both accurate and explainable



Motivation

- Rapid increase in global healthcare costs
- Lack of trust in automated prediction systems
- Users need to understand that:
  - Why is this cost predicted for me?



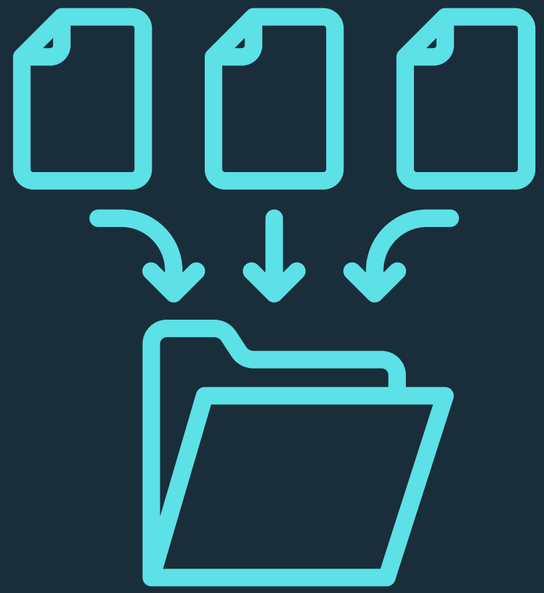
Objectives

- Predict medical insurance costs accurately
- Compare different machine learning models
- Apply Explainable AI techniques (SHAP, LIME)

Our Model Provide:

- Global explanations
- Individual-level explanations

Dataset Overview



Content:

- Source: Kaggle datasets
- Total samples: ~5000

Features:

- Age
- BMI
- Smoking status
- Region
- Number of children
- Sex

Target Variable:

- Medical insurance charges

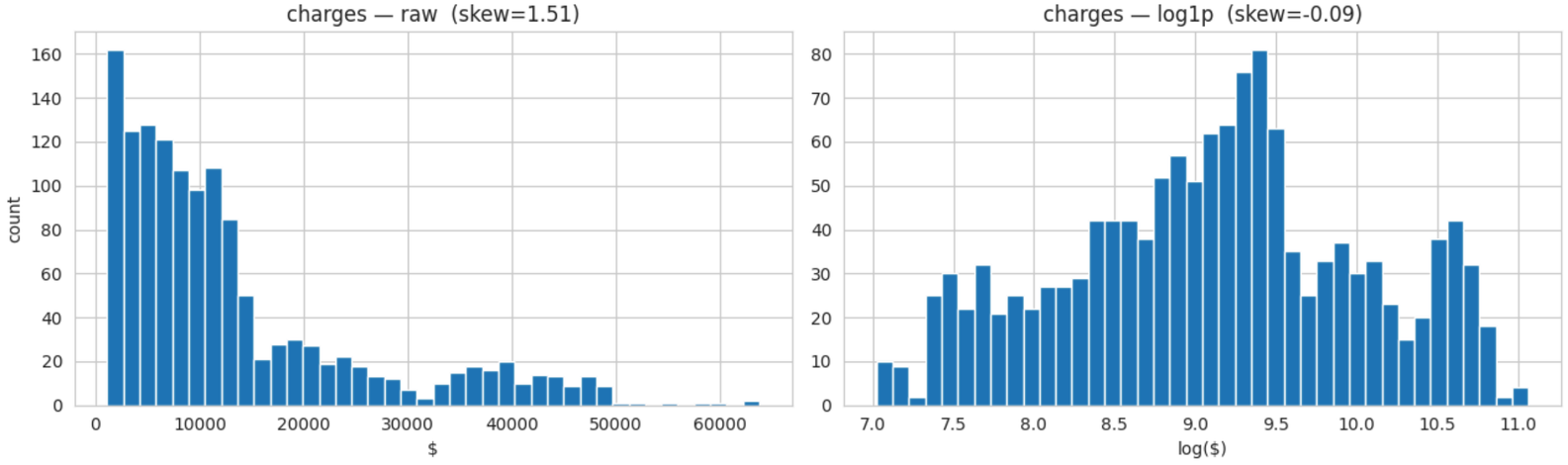


Data Preprocessing

- Data Cleaning & Inspection
  - Checked for missing values and inconsistencies
  - Performed initial data exploration
- Categorical EncodingApplied Label Encoding:
  - Sex, Smoker, Region
  - Suitable for tree-based models

Target Transformation

- High skewness in charges
- Applied log1p transformation



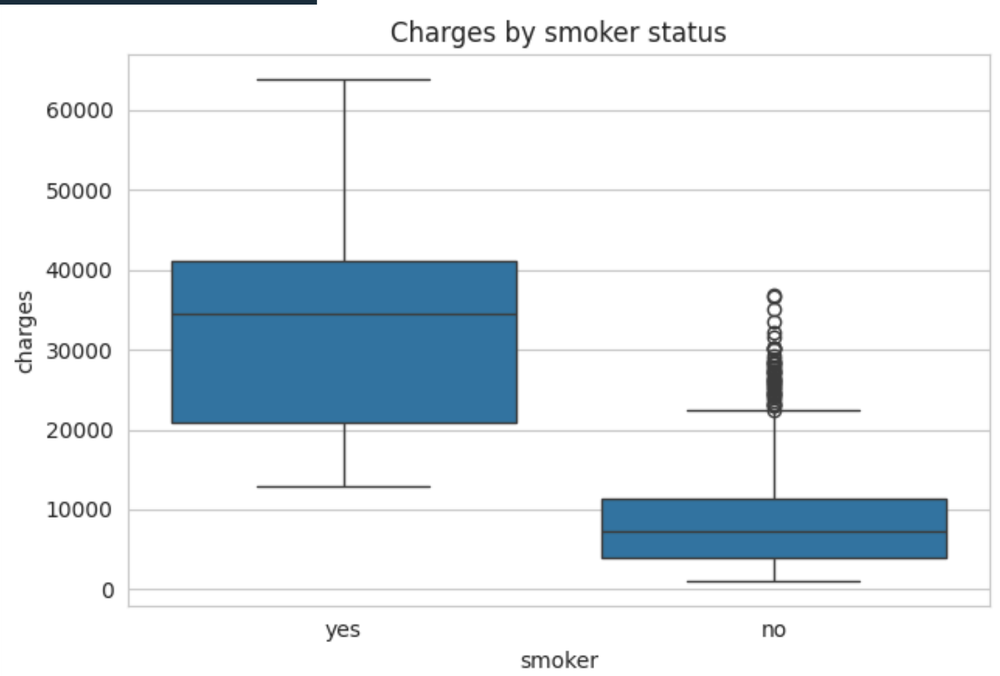
After Preprocessing:

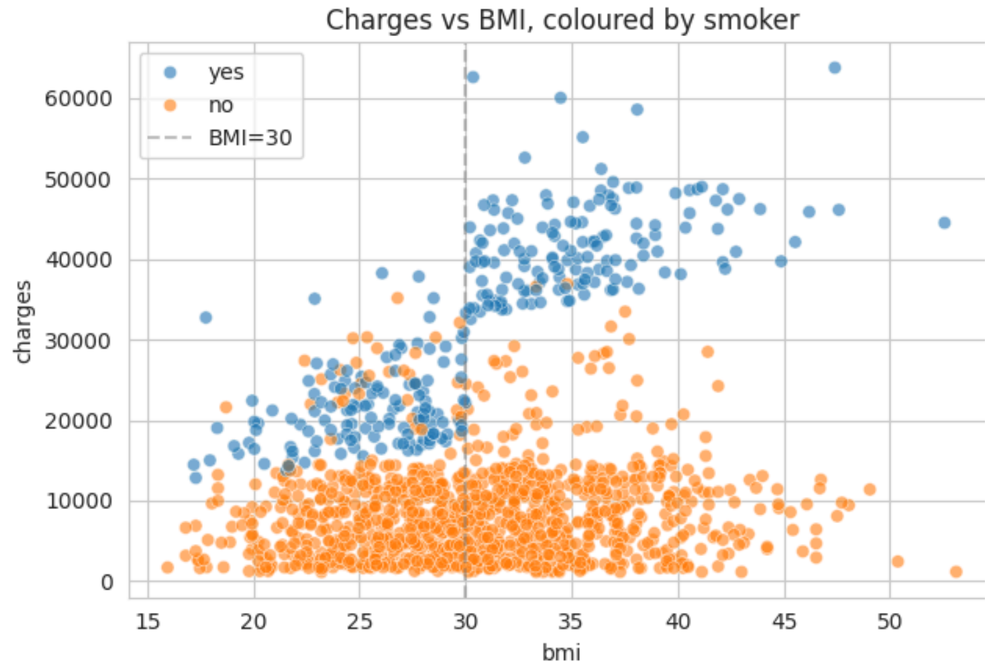
- Improves model performance and stability
- Reduces effect of extreme values

Feature Engineering

- Key Interaction Identified
  - Smoking × BMI relationship
  - Insight from AnalysisBMI has low impact for non-smokers
  - Strong impact for smokers
  - Engineered FeaturesSmoking × BMI interaction term
  - Obese smoker indicator

- Impact on Model
  - Captures non-linear and exponential cost patterns
  - Significantly improves prediction accuracy





Models Used

Baseline Models

- Linear Regression
- Decision Tree

Ensemble Models

- Random Forest
- XGBoost
- LightGBM

Deep Learning Model

- Artificial Neural Network

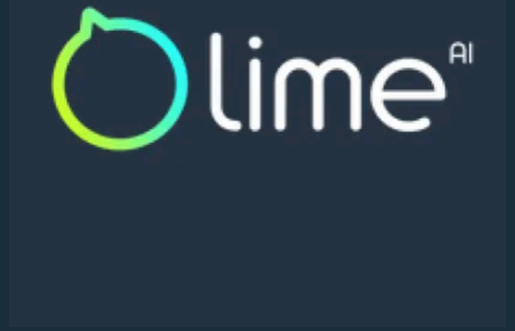


Training Setup

- All models trained on log1p(charges)
- Predictions converted back using expm1

  Ensures stable training and fair evaluation





Explainable AI (XAI)

- SHAP:
  - Measures feature contribution
  - Provides global & local explanations
- LIME:
  - Explains individual predictions
  - Uses local approximation

- Converts model from:

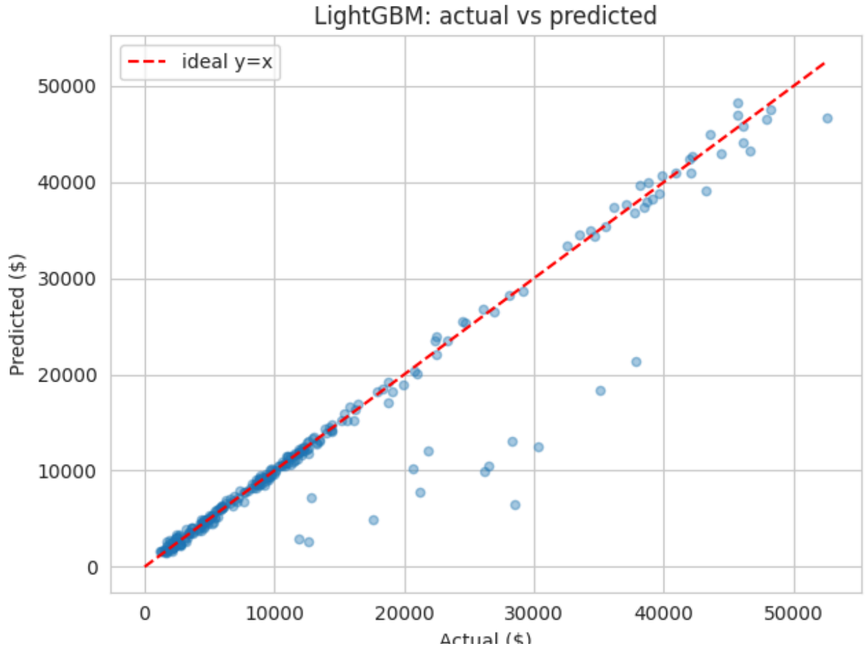
  Black Box → Explainable System

Model Performance

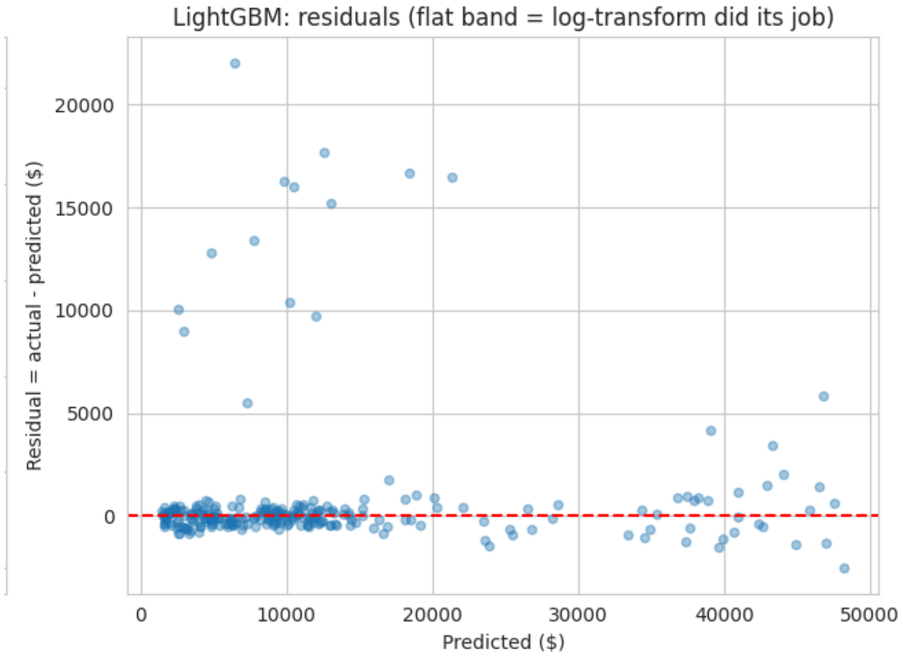
Models	MAE	RMSE	R2	MAPE
LightGBM	1132.73	3341.64	0.92	8.75
XGBoost	1297.65	3521.68	0.91	8.73
Random Forest	1735.88	3460.79	0.92	19.15
Decision Tree	1786.06	3304.73	0.92	16.25
ANN	3311.9	5366.55	0.8	24.03
Linear Regression	3570.15	7543.87	0.61	23.76



Best Performing Model: LightGBM



Actual vs Predicted values closely aligned



Residuals centered around zero

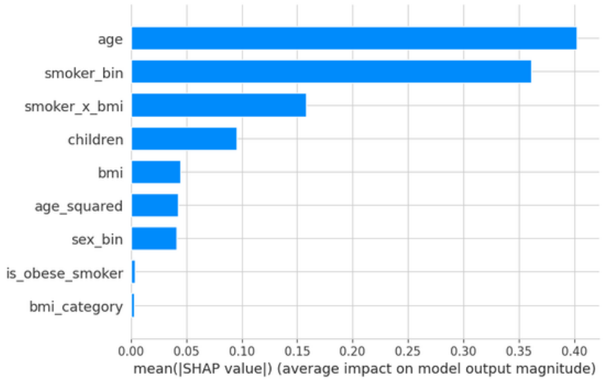
Result Visualization

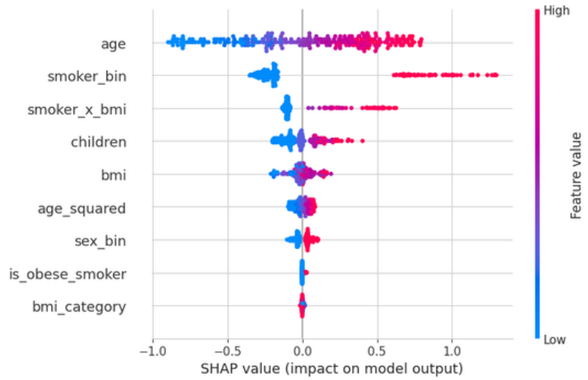
The graph Indicates:

- High prediction accuracy
- Stable model performance
- Low error variance

Explainability Results

SHAP (Global & Local Insights)





- Smoking = most influential feature
- Age and BMI also highly significant
- Strong interaction:

  Smoking × BMI increases cost sharply

Provides both:

- Global feature importance
- Individual prediction breakdown

LIME (Local Explanation)

Explains predictions for specific patients

High-cost patient ($52K):

- Smoking → strongest positive impact
- Age and Smoking × BMI → major cost drivers

```
--- HIGHEST cost test patient (actual=$52,591) -----
smoker_bin > 0.00                                +1.0817
age > 51.00                                      +0.7437
smoker_x_bmi > 0.00                              +0.4487
children <= 0.00                                 -0.1824
30.50 < bmi <= 34.80                             +0.0937
age_squared > 2601.00                            +0.0847
sex_bin <= 0.00                                  -0.0774
is_obese_smoker > 0.00                           +0.0283
```

Low-cost patient ($1.1K):

- Non-smoker → strongest negative impact
- Young age → lowers predicted cost

```
---- LOWEST  cost test patient (actual=$1,132) -----
smoker_bin <= 0.00                               -1.0864
age <= 26.00                                     -0.8287
smoker_x_bmi <= 0.00                             -0.4688
children <= 0.00                                 -0.1966
sex_bin <= 0.00                                  -0.1015
age_squared <= 676.00                            -0.0866
is_obese_smoker <= 0.00                          -0.0324
26.41 < bmi <= 30.50                             -0.0210
```

Conclusion

Summarising our system:

- Developed accurate cost prediction system
- LightGBM achieved best performance
- SHAP & LIME improved transparency

Our System benefits:

- Insurance companies
- Patients understanding cost drivers

Future Works

- Use real-world clinical data
- Include disease and genetic factors
- Deploy as web-based application

Future Goal: Real-time, user-friendly prediction system



Thank You
so much for your
Attention!

Floor is open to
Questions!

